

Complex Variables & Linear Algebra (BMAT201L)

Module 1

Analytic Functions

Dr. T. Phaneendra

July 19, 2022

1 Differentiability and Analyticity

Let $S \subset \mathbb{C}$ be a path-connected open set and $z_0 \in S$. A mapping $f : S \rightarrow \mathbb{C}$ is said to be *differentiable* at z_0 , if $f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$, provided this limit exists, and $f'(z_0)$ is called the *derivative* of f at z_0 .

Limit Test for Nonexistence of the derivative

If $L = \frac{f(z) - f(z_0)}{z - z_0}$ is different along different paths of approach as $z \rightarrow z_0$, then the derivative $f'(z_0)$ of f does not exist at z_0 .

Example 1.1. Consider $f(z) = f(x + iy) = x$ for all $z \in S$. Let

$$L = \frac{f(z) - f(0)}{z - 0} = \frac{x}{x + iy}.$$

Along the real axis, $y = 0$ and $x \rightarrow 0$ as $z \rightarrow 0$. Thus $L = x/x = 1$, while along the imaginary axis, $x = 0$ and $y \rightarrow 0$ as $z \rightarrow 0$. Thus $L = x/iy = 0$. That is, L is different along different paths of approach as $z \rightarrow 0$. Therefore, $f'(0)$ does not exist, and f is not differentiable at $z = 0$.

Cauchy-Riemann equations

Theorem 1.1. Consider $f(z) = u(z) + iv(z)$ and $z_0 = (x_0, y_0)$. If $f'(z_0)$ exists, then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \text{ at } (x_0, y_0). \quad (1.1)$$

Conditions (1.1) are called the *Cauchy-Riemann* equations.

Remark 1.1. If the Cauchy-Riemann equations are not satisfied at a point, f cannot be differentiable at that point. Thus the conditions (1.1) are *necessary* for a function $f(z)$ to be differentiable at a point z_0 .

Example 1.2. The complex function $f(z) = x - iy$ is nowhere differentiable.

Exercise 1.1. Show that $f(z) = \operatorname{Re}(z)$ is everywhere differentiable as a real variable function, but nowhere differentiable as a complex variable function.

Remark 1.2. Cauchy-Riemann equations (1.1) are not sufficient for the differentiability of f at a point z_0 . That is, a complex function f may satisfy the Cauchy-Riemann equations at a point, without being differentiable at that point.

Example 1.3. Consider

$$f(z) = \begin{cases} \frac{\bar{z}^2}{z} = \frac{\bar{z}^3}{|z|^2} = \left(\frac{x^3 - 3xy^2}{x^2 + y^2} \right) + i \left(\frac{y^3 - 3x^2y}{x^2 + y^2} \right) & \text{if } z \neq 0 \\ 0, & \text{if } z = 0. \end{cases}$$

Then

$$\frac{\partial u}{\partial x}(0,0) = \lim_{x \rightarrow 0} \frac{u(x,0) - u(0,0)}{x-0} = \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{x^3-0}{x^2+0} \right) = 1.$$

Similarly, $\frac{\partial u}{\partial y}(0,0) = 0$, $\frac{\partial v}{\partial x} = 0$ and $\frac{\partial v}{\partial y}(0,0) = 1$. Thus the Cauchy-Riemann equations hold good at the origin. Write

$$L = \frac{f(z) - f(0)}{z-0} = \frac{z^2}{z^2}.$$

Along the line $y = mx$: $z = (1 + im)x$ and $L = \frac{(1-im)^2}{(1+im)^2}$. Thus L is different for different m -values, that is L is different along different linear paths $y = mx$ of approach as $z \rightarrow 0$. Hence, f is not differentiable at 0.

Exercise 1.2. Show that each of the following functions $f(z)$ satisfies the Cauchy Riemann equations at the origin, but $f'(0)$ does not exist:

$$(a) f(z) = \begin{cases} \frac{xy^2(x+iy)}{x^2+y^4}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

$$(b) f(z) = \begin{cases} \frac{x^2y^5(x+iy)}{x^4+y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

$$(c) f(z) = \begin{cases} \frac{x^3y(y-ix)}{x^6+y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

$$(d) f(z) = \begin{cases} \frac{x^3-y^3}{x^2+y^2} + i \left(\frac{x^3+y^3}{x^2+y^2} \right), & z \neq 0 \\ 0, & z = 0 \end{cases}$$

$$(e) f(z) = \begin{cases} \frac{xy}{x^2+y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

$$(f) f(z) = \begin{cases} \frac{x^3y^2}{(x^2+y^2)^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

Theorem 1.2. Consider $f(z) = u(z) + iv(z)$ and $z_0 = (x_0, y_0) \in \mathbb{C}$. Suppose that

- (a) the partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous at (x_0, y_0) , and
- (b) satisfy the Cauchy-Riemann equations: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.

Then f will be differentiable at z_0 , and its derivative is given by

$$f'(z_0) = \frac{\partial u}{\partial x}(x_0, y_0) + i \frac{\partial v}{\partial x}(x_0, y_0). \quad (1.2)$$

Cauchy-Riemann Equations in Polar Form

If $f(z) = u(r, \theta) + iv(r, \theta)$ is differentiable at $z = re^{i\theta} \neq 0$, then

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}, \text{ and} \quad (1.3)$$

$$f'(z) = (\cos \theta - i \sin \theta) \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right) = \frac{\cos \theta - i \sin \theta}{r} \left(\frac{\partial v}{\partial \theta} - i \frac{\partial u}{\partial \theta} \right) \quad (1.4)$$

Example 1.4. Using the polar Cauchy-Riemann equations, let us find the derivative of $f(z) = z^n$, where $n \geq 2$ at $z \neq 0$.

Solution. Let $z = re^{i\theta} \neq 0$. Then $r \neq 0$, and

$$f(z) = z^n = \left(re^{i\theta} \right)^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta) = u + iv$$

so that $u = r^n \cos n\theta$ and $v = r^n \sin n\theta$, and

$$\begin{aligned} \frac{\partial u}{\partial r} &= nr^{n-1} \cos n\theta, \frac{\partial u}{\partial \theta} = -nr^n \sin n\theta \\ \frac{\partial v}{\partial r} &= nr^{n-1} \sin n\theta, \frac{\partial v}{\partial \theta} = nr^n \cos n\theta. \end{aligned}$$

We find that

$$\begin{aligned} f'(z) &= \frac{\cos \theta - i \sin \theta}{r} \left(\frac{\partial v}{\partial \theta} - i \frac{\partial u}{\partial \theta} \right) \\ &= \frac{\cos \theta - i \sin \theta}{r} (nr^n \cos n\theta + inr^n \sin n\theta) \\ &= nr^{n-1} [(\cos n\theta \cos \theta + \sin n\theta \sin \theta) + i(\sin n\theta \cos \theta - \cos n\theta \sin \theta)] \\ &= nr^{n-1} [\cos(n-1)\theta + i \sin(n-1)\theta] \\ &= nr^{n-1} e^{i(n-1)\theta} = nz^{n-1}, z \neq 0. \end{aligned}$$

Analytic Function

A function f is said to be *analytic* at a point $z_0 \in S$, if it is differentiable on some neighborhood of z_0 . A function $w = f(z)$ is *analytic* on S , if and only if it is analytic at every point of S .

Remark 1.3. Everywhere differentiable functions are analytic everywhere on $\mathbb{C}$. Conversely, everywhere analytic functions are everywhere differentiable. A function which is analytic throughout the complex plane $\mathbb{C}$ is called an *entire* function.

Example 1.5. The exponential mapping e^z is entire

Solution. We have

$$e^z = e^{x+iy} = e^x \cdot e^{iy} = e^x(\cos y + i \sin y) = u + iv, \quad (1.5)$$

so that $u = e^x \cos y$ and $v = e^x \sin y$. Then $\frac{\partial u}{\partial x} = e^x \cos y$, $\frac{\partial u}{\partial y} = -e^x \sin y$, $\frac{\partial v}{\partial x} = e^x \sin y$, $\frac{\partial v}{\partial y} = e^x \cos y$. Thus at every point $z = (x, y)$ the Cauchy-Reimann equations (1.1) hold good. Also, these first order partial derivatives are known to be continuous in the entire complex plane. Thus, by Theorem 1.2, $f = u + iv$ is analytic throughout the complex plane and hence entire.

Example 1.6. The trigonometric sine $f(z) = \sin z$ is analytic in the entire complex plane

Solution. We have

$$\begin{aligned} \sin z &= \sin(x + iy) = \sin x \cos(iy) + \cos x \sin(iy) \\ &= \sin x \cosh y + i \cos x \sinh y = u + iv \end{aligned}$$

so that $u = \sin x \cosh y$ and $v = \cos x \sinh y$. Then $\frac{\partial u}{\partial x} = \cos x \cosh y$, $\frac{\partial u}{\partial y} = \sin x \sinh y$, $\frac{\partial v}{\partial x} = -\sin x \sinh y$, $\frac{\partial v}{\partial y} = \cos x \cosh y$. Thus at every point $z = (x, y)$ the Cauchy-Reimann equations (1.1) hold good. Also, these first order partial derivatives are known to be continuous in the entire complex plane. Thus, by Theorem 1.2, $f = u + iv$ is analytic throughout the complex plane. Also,

$$\begin{aligned} f'(z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \cos x \cosh y + i[-\sin x \sinh y] \\ &= \cos x \cos(iy) - \sin x \sin(iy) = \cos(x + iy) = \cos z \end{aligned}$$

Example 1.7. The trigonometric cosine $f(z) = \cos z$ is an entire function

Solution. We have

$$\begin{aligned} \cos z &= \cos(x + iy) = \cos x \cos(iy) - \sin x \sin(iy) \\ &= \cos x \cosh y - i \sin x \sinh y = u + iv. \end{aligned}$$

Then proceed as in the previous lines.

Example 1.8. Identify the real and imaginary parts of $f(z) = e^{z^2}$, and show that f is everywhere differentiable

Let $z = x + iy$. Then $z^2 = (x + iy)^2 = x^2 - y^2 + i.2xy$, and

$$\begin{aligned} e^{z^2} &= e^{x^2 - y^2 + i.2xy} = e^{x^2 - y^2} \cdot e^{i.2xy} \\ &= e^{x^2 - y^2} (\cos 2xy + i \sin 2xy) = u + iv \end{aligned}$$

so that $u = e^{x^2-y^2} \cos 2xy$ and $v = e^{x^2-y^2} \sin 2xy$ are real and imaginary parts of f . Since f is a composition of the entire functions e^z and z^2 , it will also be analytic, and by the chain rule of differentiation, we have

$$f'(z) = \frac{d}{dz} (e^{z^2}) = e^{z^2} (2z) = 2ze^{z^2}$$

Example 1.9. Identify the real and imaginary parts of $f(z) = e^{e^z}$, and hence show that f is everywhere differentiable

We see that $e^{e^z} = e^{e^x \cos y} [\cos(e^x \sin y) + i \sin(e^x \sin y)] = u + iv$, so that $u = e^{e^x \cos y} \cos(e^x \sin y)$ and $v = e^{e^x \cos y} \sin(e^x \sin y)$, and $f'(z) = e^z e^{e^z}$

Example 1.10. Show that $f(z) = e^{1/z}$ is analytic at all $z \neq 0$, and hence find its derivative

For $z \neq 0$, $e^{1/z} = e^{x/(x^2+y^2)} \left[\cos\left(\frac{y}{x^2+y^2}\right) - i \sin\left(\frac{y}{x^2+y^2}\right) \right] = u + iv$ so that $u = e^{x/(x^2+y^2)} \cos\left(\frac{y}{x^2+y^2}\right)$ and $v = -e^{x/(x^2+y^2)} \sin\left(\frac{y}{x^2+y^2}\right)$

Example 1.11. The principal branch

$$F(z) = \log \sqrt{x^2 + y^2} + i \tan^{-1}(y/x) \quad (1.6)$$

of the complex logarithm $f(z) = \log z$, is analytic at all points $z = x + iy \in \mathbb{C}$ with

$$\sqrt{x^2 + y^2} > 0 \text{ and } -\pi < \tan^{-1}(y/x) < \pi. \quad (1.7)$$

Solution. Let $z = re^{i\theta}$ with $r > 0$ and $-\pi < \theta < \pi$. Then

$$F(z) = \log \sqrt{x^2 + y^2} + i \tan^{-1}(y/x) = \log r + i\theta = u(r, \theta) + iv(r, \theta),$$

so that real and imaginary parts of F are $u(r, \theta) = \log r$ and $v(r, \theta) = \theta$ respectively. We find that

$$\frac{\partial u}{\partial r} = \frac{1}{r}, \frac{\partial u}{\partial \theta} = 0, \frac{\partial v}{\partial r} = 0, \frac{\partial v}{\partial \theta} = 1.$$

Thus u and v satisfy the Cauchy-Reimann equations (1.3). Since u , v and their first partial derivatives are continuous at all points $z = x + iy \in \mathbb{C}$ in the domain (1.7), it follows that F is analytic in this domain. Also, the derivative of F is given by

$$f'(z) = (\cos \theta - i \sin \theta) \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right) = e^{-i\theta} \left(\frac{1}{r} + i \cdot 0 \right) = 1/re^{i\theta} = 1/z$$

Exercise 1.3. Identify the real and imaginary parts of $f(z) = (z^2 - 2)e^{-z}$, and show that f is entire. Then find $f'(z)$.

Exercise 1.4. Explain why $f(z) = 2z^2 - 3 - ze^z + e^{-z}$ and $g(z) = 3z^2 + 5z - 6i$ are

entire

Exercise 1.5. Explain why $f(z) = x + \sin x \cosh y + i(y + \cos x \sinh y)$ is analytic at every point of the complex plane

Exercise 1.6. Show that each of the following functions $f(z)$ is entire, and find the derivative at each point z of the complex plane:

(a) $z^3 = x^3 - 3xy^2 + i(3x^2y - y^3)$

(b) $4x^2 + 5x - 4y^2 + 9 + i(8xy + 5y - 1)$

Exercise 1.7. Find the choice of the constants (*mentioned in brackets*) such that each of the following functions is entire:

(a) $f(z) = x + ay + i(bx + cy)$, (a, b, c)

(b) $f(z) = (ax + y) + i(3y - bx)$, (a, b)

(c) $f(z) = e^x[\cos ay + i \sin(y + b)] + c$, (a, b, c)

(d) $f(z) = x^2 + axy + by^2 + i(cx^2 + dxy + y^2)$, (a, b, c, d)

(e) $f(z) = (3x - y + 5) + i(ax + b - 3)$, (a, b)

Exercise 1.8. Show that each of the following functions $f(z)$ is analytic in an appropriate domain, and find its derivative in the domain:

(a) $\frac{x-1}{(x-1)^2+y^2} - i\frac{y}{(x-1)^2+y^2}$

(b) $\frac{x^3+xy^2+x}{x^2+y^2} + i\frac{x^2y+y^3-y}{x^2+y^2}$

Remark 1.4. Analyticity at a point implies the differentiability at a point, but the converse is not true. In fact, there are functions which are differentiable at a point but are nowhere analytic

Example 1.12. Consider $f(z) = |z|^2 = x^2 + y^2$ for all $z = (x, y) \in \mathbb{C}$. Then $u_x \equiv \frac{\partial u}{\partial x} = 2x$, $u_y \equiv \frac{\partial u}{\partial y} = 2y$, $v_x \equiv \frac{\partial v}{\partial x} = 0$, $v_y \equiv \frac{\partial v}{\partial y} = 0$, which are continuous functions and $u_x = v_y$, $v_x = -u_y$ only if $x = 0 = y$. That is, the Cauchy-Riemann equations are satisfied at $z = (0, 0)$ only. Thus, by Theorem 1.2, f is differentiable only at $z = 0$. Since the Cauchy-Riemann equations are not satisfied at $z \neq (0, 0)$, there exists no neighborhood of $(0, 0)$, in which f is differentiable, except at $(0, 0)$. That is, f is not analytic at 0.

Since the Cauchy-Riemann equations are not satisfied at the points $z \neq 0$, f is not differentiable and hence not analytic at $z = 0$. Thus f is nowhere analytic.

Example 1.13. If $f(z) = x^2y^2$ for all $z = (x, y) \in \mathbb{C}$, we see that

$$u_x \equiv \frac{\partial u}{\partial x} = 2xy^2, u_y \equiv \frac{\partial u}{\partial y} = 2x^2y, v_x \equiv \frac{\partial v}{\partial x} = 0, v_y \equiv \frac{\partial v}{\partial y} = 0,$$

which are continuous functions and

$$u_x = v_y, v_x = -u_y \quad \text{only if } x = 0 \text{ or } y = 0.$$

That is, the Cauchy-Riemann equations are satisfied at all points on the coordinate axes. Then by Theorem 1.2, f will be differentiable on the coordinate axes. But there is no neighborhood of any of the points on the axes, on which f is differentiable. Hence it is nowhere analytic.

Exercise 1.9. Show that $f(z) = e^{i\bar{z}}$ and $g(z) = 2x + ixy^2$ are nowhere differentiable and hence nowhere analytic

Exercise 1.10 (HOT). Show that

$$\begin{aligned} f(z) &= \cos x - i \sinh y \\ g(z) &= (\bar{z} + 1)^3 - 3\bar{z} \end{aligned}$$

are nowhere analytic

Exercise 1.11. Show that each of the following functions is differentiable only at the origin but nowhere analytic:

- (a) $z^2\bar{z} = z\bar{z}^2, z\operatorname{Im}z$
- (b) $|z|^2 = x^2 + y^2, x^2 - ixy$

Exercise 1.12. Show that each of the following functions is nowhere analytic but is differentiable along the indicated curve(s)

- (a) $f(z) = x^2 + y^2 + i2xy$, the real axis
- (b) $f(z) = 3x^2y^2 - i6x^2y^2$, the coordinate axes
- (c) $f(z) = x^3 + 3xy^2 - x + i(y^3 + 3x^2 - y)$, the coordinate axes

Exercise 1.13 (HOT). If $f(z) = \begin{cases} e^{-1/z^4}, & z \neq 0 \\ 0, & z = 0, \end{cases}$ show that the Cauchy-Riemann equations are satisfied at every point of the complex plane but f is analytic only at all $z \neq 0$.

Exercise 1.14 (HOT). Prove or disprove the following statements:

- (a) Everywhere differentiable functions are analytic everywhere

- (b) If $f = u + iv$ satisfies the Cauchy-Riemann equations at every point of the complex plane, then it will be differentiable at at least one point of the complex plane
- (c) Nowhere analytic function is nowhere differentiable
- (d) If f is differentiable at a point of the complex plane, then it will be analytic at that point
- (e) If $f = u + iv$ is analytic, then $f = v + iu$ is analytic

Exercise 1.15. If $f = u + iv$ is differentiable at a point, compute the Jacobian $J\left(\frac{u,v}{x,y}\right)$ in terms of the derivative $f'(z)$

Exercise 1.16 (HOT). If $f(z)$ is differentiable at a point z_0 , show that

$$\left(\frac{\partial f}{\partial z} + \frac{\partial f}{\partial \bar{z}}\right) + \left(\frac{\partial f}{\partial z} - \frac{\partial f}{\partial \bar{z}}\right) = \frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} = 0 \text{ at } z_0$$

Exercise 1.17 (HOT). Show that $f(z) = \sin(\bar{z})$ is nowhere differentiable and hence nowhere analytic

2 Applications

Theorem 2.1. If $f(z)$ is an analytic with constant modulus (that is, $|f(z)| = \text{constant}$) on a domain S , then $f(z)$ will also be constant on S .

Theorem 2.2. If $f(z)$ is an analytic with $f'(z) = 0$ on a domain S , then $f(z)$ will be constant on S .

Harmonic function: A function $\psi(x, y)$ which has continuous partial derivatives of second order on a domain S is said to be *harmonic*, if it satisfies the Laplace's equation $\nabla^2 \psi \equiv \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$ on S .

Theorem 2.3. Let $f(z) = u(x, y) + iv(x, y)$ be analytic on a domain S . Then the real and imaginary parts $u \equiv u(x, y)$ and $v \equiv v(x, y)$ will be harmonic on S , that is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \text{ and } \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \text{ on } S.$$

Remark 2.1. If $f = u + iv$ is analytic, we say that the imaginary part v is a harmonic conjugate of the real part u of f . Note that v is a harmonic conjugate of u if and only if u is a harmonic conjugate of $-v$. This follows from the observation that $if = i(u + iv) = -v + iu$ is analytic whenever $f = u + iv$ is analytic. This property is referred to as the *antisymmetry* of harmonic conjugates.

Finding a Harmonic Conjugate

Let f be an analytic function with the real part u . Then u is harmonic. The imaginary part v of f is the conjugate harmonic of u , and is determined by integrating the total derivative

$$dv = \left(\frac{\partial v}{\partial x}\right) dx + \left(\frac{\partial v}{\partial y}\right) dy = -\left(\frac{\partial u}{\partial y}\right) dx + \left(\frac{\partial u}{\partial x}\right) dy.$$

Thus the short-cut formula is

$$v \equiv v(x, y) = - \int_{y-\text{constant}} \left(\frac{\partial u}{\partial y}\right) dx + \int_{x-\text{free}} \left(\frac{\partial u}{\partial x}\right) dy + \text{constant} \quad (2.1)$$

Exercise 2.1. Given a harmonic function $u \equiv u(x, y)$, determine its harmonic conjugate $v \equiv v(x, y)$ and the analytic function $f(z) = u + iv$:

- (a) $\cos x \cosh y$
- (b) $y + e^x \cos y, e^{2xy} \sin(x^2 - y^2), e^{x^2 - y^2} \cos(2xy)$
- (c) $x^2 - y^2, 4xy^3 - 4x^3y + x$
- (d) $\frac{x^2 - y^2}{(x^2 + y^2)^2}, x^2 - y^2 + \frac{x}{x^2 + y^2}$
- (e) $\frac{\sin 2x}{\cos 2x + \cosh 2y}, e^x(x \cos y - y \sin y)$

Remark 2.2. Let f be an analytic function with the imaginary part v . Then v is harmonic. The real part u of f is determined by the short-cut formula

$$u \equiv u(x, y) = \int_{y-\text{constant}} \left(\frac{\partial v}{\partial y}\right) dx - \int_{x-\text{free}} \left(\frac{\partial v}{\partial x}\right) dy + \text{constant} \quad (2.2)$$

Milne-Thomson method: Given the harmonic function $u(x, y)$, to find the corresponding analytic function f whose real part is u , first we write

$$f'(z) = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}.$$

We replace x by z and y by 0 in this and then integrate w. r. t. z .

Exercise 2.2. Without finding the real parts, find the analytic functions whose imaginary parts are $e^x(x \sin y + y \cos y)$ and $\frac{x-y}{x^2+y^2}$, respectively.

Exercise 2.3 (HOT). Determine the analytic function $f(z) = u + iv$ where u and v satisfy the following relations:

- (a) $u - v = \frac{\cos x + \sin x - e^{-y}}{2(\cos x - \cosh y)}$
- (b) $u + v = \frac{x-y}{x^2 + 4xy + y^2}$

Theorem 2.4 (Analytic Function and Orthogonal Curves). Suppose that $f(z) = u(x, y) + iv(x, y)$ is analytic at a point P . Then the curves $u(x, y) = a$ and $v(x, y) = b$ are orthogonal at P .

Flow Problems: Consider the irrotational motion of a frictionless, incompressible fluid moving in planes parallel to the xy -plane (that is, planar flow). Suppose that $\mathbf{F}(x, y) = f_1\mathbf{i} + f_2\mathbf{j}$ is the two dimensional velocity field of a fluid particle. Then $\nabla \times \mathbf{F} = 0$ so that $\mathbf{F} = \text{grad } \phi$ for some scalar potential $\phi(x, y)$. Since the flow is incompressible, $\nabla \cdot \mathbf{F} = 0$ or $\nabla \cdot \text{grad } \phi = \nabla^2 \phi = 0$, that is ϕ is harmonic. The scalar potential ϕ is called a *velocity potential*. It follows that there exists a harmonic conjugate $\psi(x, y)$ of ϕ such that $f(z) = \phi(x, y) + i\psi(x, y)$ is analytic, and the harmonic conjugate ψ is called the *stream function*. Since f is analytic, the level curves $\phi(x, y) = c_1$ and $\psi(x, y) = c_2$ are orthogonal. The level curves corresponding to the velocity potential ϕ are known as *equipotential* curves, while the level curves corresponding to the stream function are called the *stream lines*. Stream lines represent the actual paths along which the fluid particles will move. The flow pattern is fully represented by the analytic function $f(z) = \phi(x, y) + i\psi(x, y)$, called the *complex potential*. In electrostatics and gravitational fields, the curves $\phi = c_1$ and $\psi = c_2$ are equipotential lines and lines of force. In steady state heat flow problems, the curves $\phi = c_1$ and $\psi = c_2$ are isotherms and lines of flow or flux lines

Exercise 2.4 (HOT). You should be little clever in giving the conclusions for the following problems:

1. If $f(z)$ is an entire function, is $\overline{f(z)}$ analytic at at least one point of the complex plane? If both $f(z)$ and $\overline{f(z)}$ are analytic, what can you say about f ?
2. If u and v are harmonic functions, can $f = u + iv$ be analytic? Justify!
3. If $f = u + iv$ is analytic where u is a harmonic conjugate of v , what can you say about f ?
4. If $f = u + iv$ is analytic, prove that $U = e^u \cos v$ and $V = e^u \sin v$ are harmonic conjugates of each other.
5. If $u(x, y)$ is a harmonic function and $v(x, y)$ is its harmonic conjugate, show that $\phi(x, y) = u(x, y)v(x, y)$ and $\psi(x, y) = u^2 - v^2$ are harmonic.